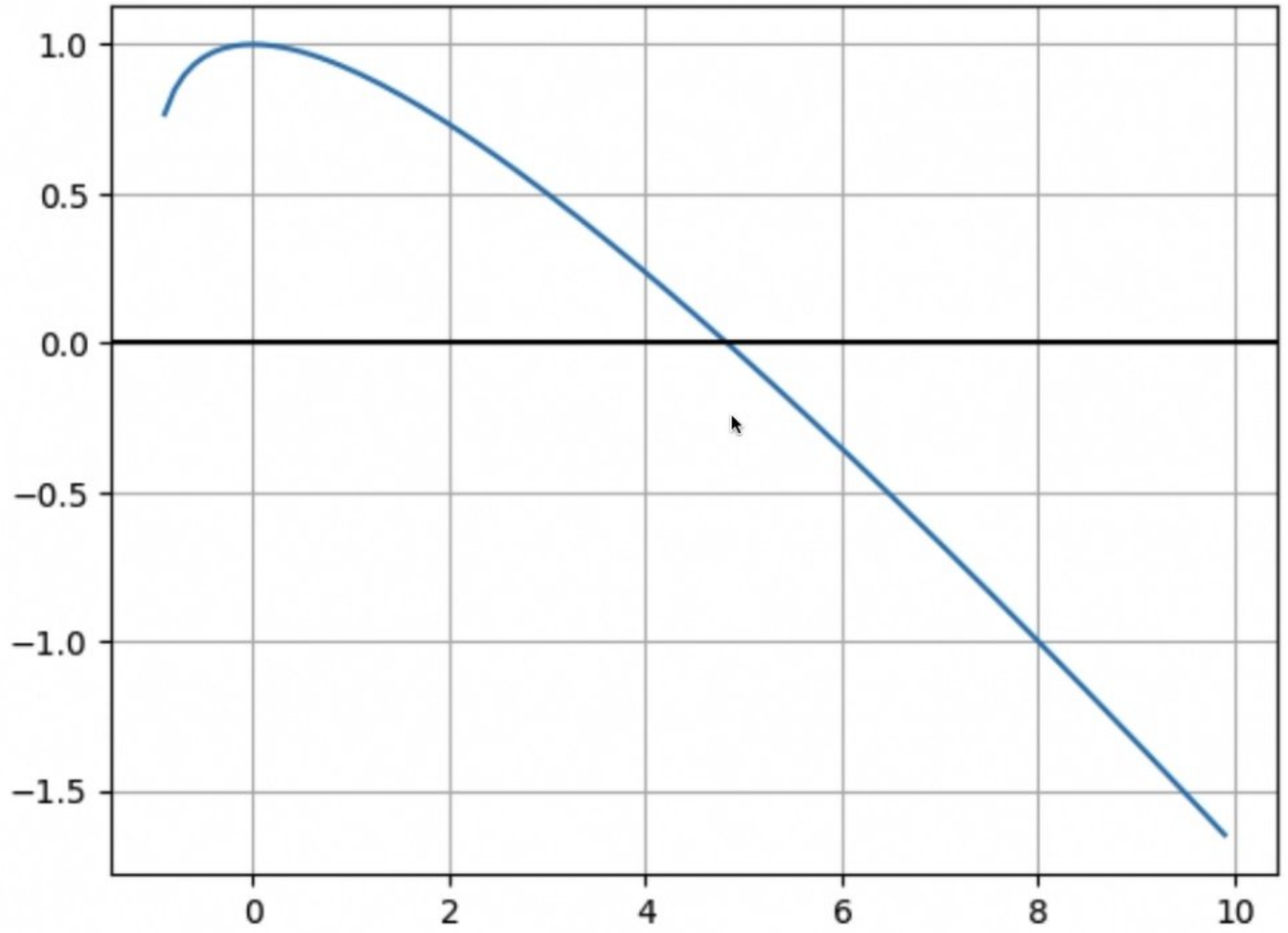
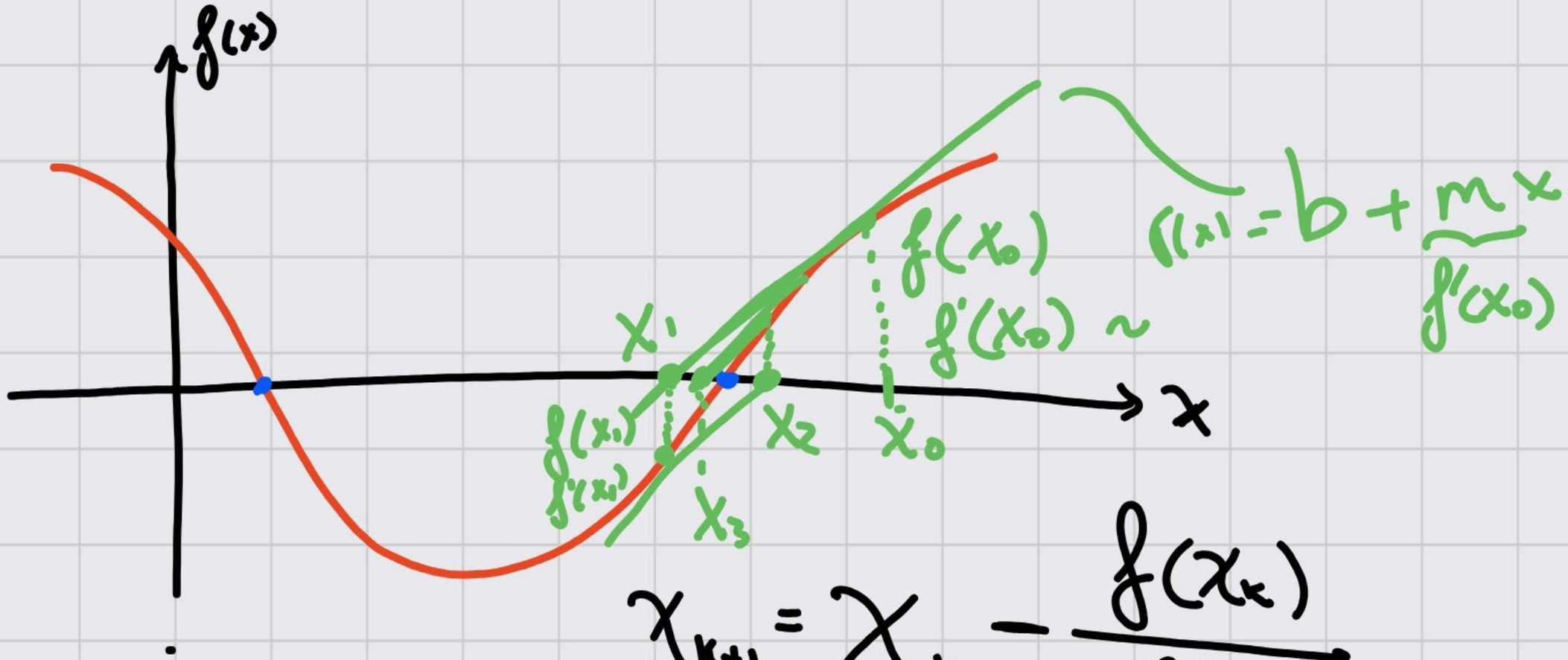
```
/tmp/ipykernel_2336/2621335232.py:2: RuntimeWarning: invalid value encountered in sqrt
  return np.sqrt(x+1) - x/2
```



Variables Terminal



$$x_1 = b + \frac{m}{f'(x_0)}x$$

$$x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)}$$

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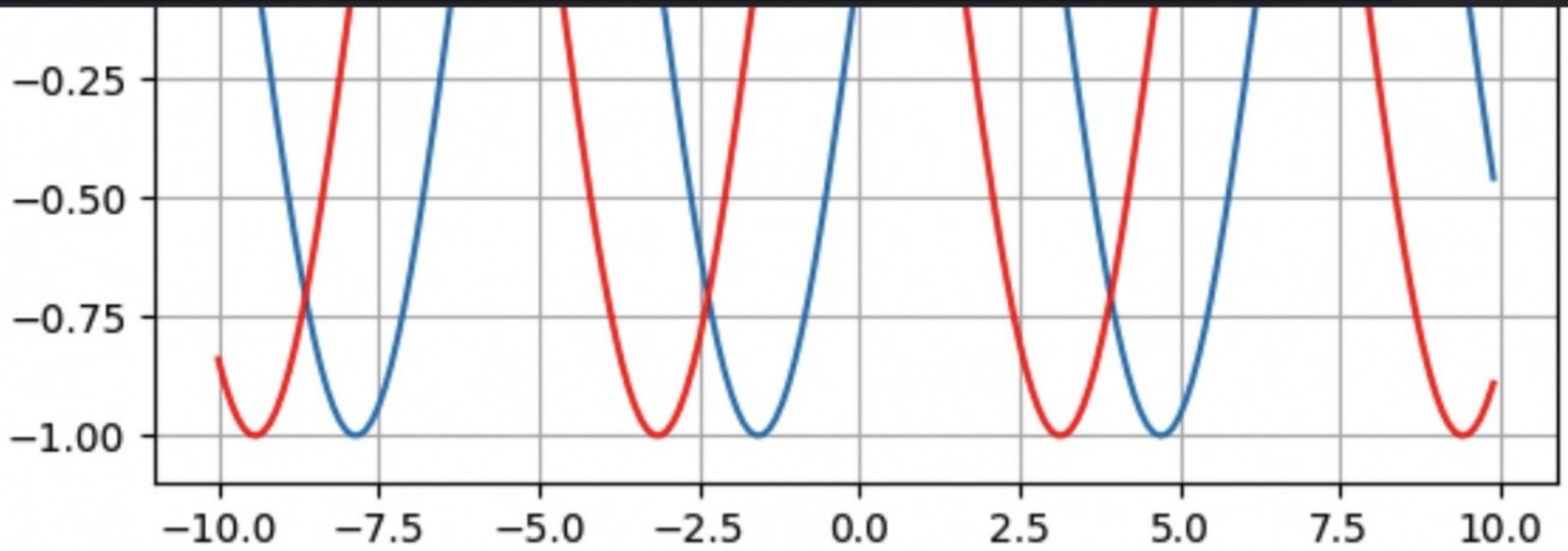
$$f(x) = \sqrt{x+1} - \frac{x}{2} = (x+1)^{1/2} - \frac{x}{2}$$

de $f(x) = 0$

$$f'(x) = \frac{1}{2}(x+1)^{-1/2} - \frac{1}{2}$$

$$= \frac{1}{2\sqrt{x+1}} - \frac{1}{2} = \frac{2 - 2\sqrt{x+1}}{2 \cdot 2\sqrt{x+1}}$$

$$= \frac{2(1-\sqrt{x+1})}{2 \cdot 2\sqrt{x+1}} = \frac{1-\sqrt{x+1}}{2\sqrt{x+1}}$$



[28]
✓ 0s

```
newton_raphson(f=f, df=df, x=2, iterations=10)
```

```
np.float64(3.141592653589793)
```

[ ]

Start coding or generate with AI.

Reg Logistica

> Modelo $P = \frac{1}{1+e^{-X\beta}} = \sigma(X\beta)$ · $X = \begin{pmatrix} 1 & x_{11} & x_{12} & \cdots \\ 1 & x_{21} & x_{22} & \cdots \\ \vdots & & & \ddots \\ & & & x_{nm} \end{pmatrix}$ $n \times (m+1)$; $y = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}$ $n \times 1$

$P = \begin{pmatrix} p_1 \\ p_2 \\ \vdots \\ p_n \end{pmatrix}$ $n \times 1$

> Funcion costo

$$l(\beta) = -\sum y_i \ln(p_i) + (1-y_i)\ln(1-p_i)$$

$y_i \in \{0, 1\}$

↳ $\operatorname{argmin}(l)$

$$\frac{\partial l}{\partial \beta_j} = X^T (y - P) \to (m+1) \times 1 = 0$$

(X^T: $(m+1) \times n$; $(y-P)$: $n \times 1$; $n \times 1$)

$$\frac{\partial^2 l}{\partial \beta_j \partial \beta_k} = X^T W X \to (m+1) \times (m+1)$$

(X^T: $m+1 \times n$; W: $n \times n$; X: $n \times m+1$)

$$W \to \operatorname{diag}(p_i(1-p_i))$$

$$x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)}$$

$$\vec{\beta}_{k+1} = \vec{\beta}_k - X^T (y-P)\left[X^T W X\right]^{-1}$$

$$\begin{pmatrix} \frac{\partial^2 l}{\partial \beta^2} & \frac{\partial^2 l}{\partial \beta_1 \partial \beta_0} & \cdots \\ \frac{\partial^2 l}{\partial \beta_0 \partial \beta_1} & \frac{\partial^2}{\partial \beta_1^2} & \cdots \\ \vdots & \vdots & \ddots \end{pmatrix}$$

```
    return sigmoid(X@beta)
```

```
beta = [[1.5],[-1.5],[-0.2]]
P(X=X_array, beta=beta)
```

```
array([[0.31002552],
       [0.26894142],
       [0.23147522],
       [0.47502081]])
```

$$P = \frac{1}{1 + e^{-X\beta}}$$

$$X\beta = \beta_0 + \beta_1 x_1^{(i)} + \beta_2 x_2^{(i)}$$

```
def W()
```

ables Terminal

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$$\frac{\partial B}{\partial B} = 0 = 2X^T Xp - 2X^T Y + 2\lambda p$$

$$\Pi = \lambda \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{pmatrix}$$

$$\beta = (X^T X + I\lambda)^{-1} X^T Y$$

cto importante. Si $X^T X$ tuviera probl

e lo hace inestable y numéricamente

Line: 2 Col: 13